

A Moving Target DDoS Defense Approach in Consortium Blockchain

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Abstract—Although consortium blockchain has an identification mechanism, the captured internal clients are potentially threatening internal blockchain nodes. Internal Distributed Denial-of-Service (DDoS) attacks threaten the specific nodes in consortium blockchain, e.g., the executor, consensus, and committer nodes. Typical attack methods may include SYN Flooding and ACK Flooding and deny normal transaction service by sending many invalid transactions and blocks. In this work, we have proposed an organization collaboration-based DDoS defense approach and a Deep Q-learning (DQN)-based Moving Target Defense (MTD) for changing attack surface of victims in consortium blockchain. On one hand, contracts are used to synchronize attack information obtained from organizations, e.g., bots' IP addresses and public keys. On the other hand, we have developed a DQN-based MTD defense mechanism for organizations to change the attack surface of victims in order to mitigate the malicious traffic, in the case of missing detections of bots. Our approach applies a multi-stage game to reflect interactions between attackers and defenders. The evaluation results have demonstrated that our approach could effectively mitigate DDoS attacks in consortium blockchain.

Index Terms—Consortium blockchain, DDoS, moving target defense, deep Q-network.

I. INTRODUCTION

TEMPORARILY, *Distributed-Denial-of-Service* (DDoS) attacks are common threats to majority of applications running on the Internet. A variety of new DDoS attacks have brought new challenges to existing DDoS defense technologies, including consortium blockchain [1], [2], [3]. Even though blockchain presents services in a decentralized manner, which implies a higher-level defense capability, the DDoS attack still is

threatening the security of blockchain [4], [5], [6], [7]. Existing Internet DDoS defense methods focus on defending large-scale malicious traffic requests on the Internet. However, due to the limited number of participating nodes, the nodes inside each organization are public for internal clients to call functions, but internal clients are probably compromised by attackers, which cannot be defended by existing Internet DDoS defense methods.

In the typical consortium blockchain, the authorized internal clients have permission to obtain the node's information and smart contracts. Most consortium blockchains (e.g., Hyperledger Fabric [8]) implement built-in rate restrictions mainly targets transaction processing speed or block generation frequency to mitigate DDoS attacks, encompassing limits on a variety of aspects, such as the number of peers and orderers, overall query capacity, concurrency thresholds, and batch timeouts. However, the traditional setting encounters a failure in defending internal DDoS attacks, since an internal DDoS attacker is able to not only launch typical DDoS attacks (e.g., SYN Flood or ACK Flood) but also compromise internal clients to implement different internal attacks at the application layer, threatening the execution, consensus, and block committing of transactions [9]. Therefore, these built-in rate limitations cannot mitigate the target DDoS attacks considered in this work. The challenge issue is that the internal attacks make various types of nodes ignore current identity authentication rather than evading Internet DDoS defense [10]. A typical internal DDoS attack scenario, for example, is that DDoS attackers may create a large number of invalid transactions (e.g., incorrect signatures) to block the execution of normal transactions for targeting executor nodes, as the blockchain network allows anyone to freely send transactions [11]. In addition, when attacking consensus nodes, the attacker can send a large number of endorsed transactions until the node's transaction pool is filled with malicious transactions. Attackers can also attack committer nodes by initiating a large number of empty blocks, unsigned blocks, or blocks containing conflict transactions to block the storage of normal blocks. The bots controlled by the attacker can rationally arrange attack resources in different organizations.

However, there are rare works focusing on the DDoS defense inside the consortium blockchain. We have investigated the typical DDoS defense approaches and analyzed the defense effect on consortium blockchain, including traffic filtering, *Software-defined Networking* (SDN) [12], *Network Function Virtualization* (NFV) [13], *Moving Target Defense* (MTD) [14], [15], [16] and collaborative defense approach [17], [18], [19], [20]. The

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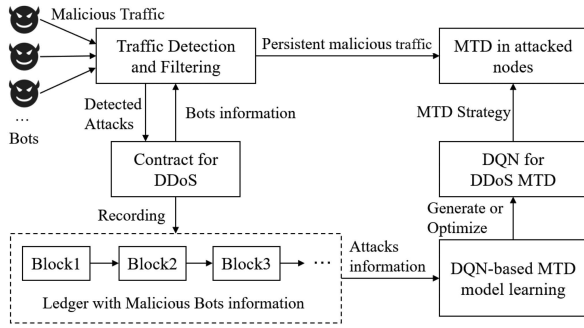


Fig. 1. The architecture of proposed DDoS defense approach.

filtering method generally performs well in most situations, but the method can hardly handle the moving attackers due to its inflexibility limitations [21]. SDN-based and NFV-based methods present higher-level flexibility but generally cause a longer latency [22]. The MTD is one of the important active defense approaches, but resource consumption and uncertainty lead to the MTD strategy not being applied in consortium blockchain directly. The collaborative defense is a suitable approach for defending against DDoS attacks inside consortium blockchain, which can not only fit the architecture of consortium blockchain but also enhance DDoS defense of single organizations.

Combining *Contract-based Collaborative Defense* with MTD is a better way to mitigate DDoS attacks in consortium blockchain. MTD avoids being attacked by malicious traffic while ensuring normal blockchain service and changes the *Attack Surface* (AS) for preventing persistent attacks. Besides, reinforcement learning [23] provides MTD strategy generation with continuous solutions that fully utilize smart contracts to collect DDoS attack information concerning attack features for every node.

In this work, we have proposed a two-stage DDoS defense approach for consortium blockchains. The architecture of proposed defense approach consists of four parts as shown in Fig. 1, including malicious traffic detection and filtering, attacks data synchronization contract, DQN-based MTD action generation, and MTD execution module. The traffic detection and filtering module filters malicious traffics by using the attack data stored in the distributed ledger through the contract. When the attacker is recorded in the ledger, the traffic is directly filtered by the filtering module. There are some persistent malicious traffics, and each node in the organization uses MTD module to change its own environment for mitigating the attacks. The MTD method in each node comes from the DQN. Attack data required by DQN model derives from the data stored in the distributed ledger.

The main contributions of this paper are as follows:

- 1) We have explored using an MTD-based mechanism to defend DDoS attacks in the consortium blockchain. In order to construct an adoptable solution, we have attempted to apply a DQN to generate MTD strategies in each organization.
- 2) We have developed a smart contract-based collaborative defense method in consortium blockchain. The proposed method allows the organizations in consortium blockchain to share information on collected malicious traffics, so

that the independent organizations have a broader scope of knowledge about attacks compared to a centralized-based defense setting.

- 3) This work utilizes blockchain smart contracts to collect and record the attack feature data in order to provide sufficient and qualified training data for DQN. Due to the collaborative defense setting, combining with smart contracts can facilitate a higher-level efficiency when adversaries appear a dynamic attacking manner.

The advantages of our proposed approach in this paper, compared to other methods, are: (i) *Focus on Consortium Blockchain Security*: Most existing work focuses on public blockchain, despite DDoS attacks posing a greater threat to consortium blockchain systems. Consortium blockchains have fewer nodes centralized within specific organizations, making them easier targets for attackers. In contrast, public blockchains have a vast, globally distributed network, requiring more resources to disrupt [24], [25]. Additionally, consortium blockchains rely on cryptographic-based consensus (e.g., PBFT or RAFT), which can be severely impacted by attacking a few key nodes. Public blockchains use more robust mechanisms (e.g., PoW or PoS) that are more resistant to such attacks. Furthermore, consortium blockchain nodes are easier to identify and target due to their limited anonymity, whereas public blockchain nodes are anonymous. Node management rules in consortium blockchains mitigate malicious attacks using Sybil accounts and other stealth strategies, serving as prerequisites for moving target defense. (ii) *Attack Data Sharing*: Existing defense methods do not effectively leverage blockchain's data sharing capabilities. Our approach combines the distributed ledger characteristics of blockchain with deep reinforcement learning to create defensive methods. The consortium blockchain provides the necessary conditions for implementing our approach and the infrastructure for data sharing, enabling each node to store a complete dataset and generate effective defense solutions using synchronized attack data.

The rest of this paper is organized by the followings. Section II presents the threat model and defines the proposed problem. Sections III and IV explain smart contract-based collaborative defense mechanism and DQN-based model, respectively. The evaluation and main findings are given in Section V. Finally, we review related work and draw conclusions in Sections VI and VII.

II. CONCEPTS AND THE DEFINITIONS

A. Threat Model

Our attack vector assumes that rational attackers can compromise internal nodes in the consortium blockchain with public information to achieve maximum profit. The goal of internal attackers launching DDoS attacks is to make the consortium blockchain unable to provide normal services, without generating direct benefits, but will appear indirectly in the form of external benefits, such as disrupting or extorting competing organizations within the consortium blockchain, or driven by economic interests. Rational attackers will terminate their attacks when the cost and benefits of launching attacks are unequal or even less. In contrast, compromised bots are authenticated to

access internal information of consortium blockchain, e.g., types of nodes, deployed contracts of different nodes, and addresses of internal nodes.

DDoS attackers can make the optimal attack strategy deriving from the information asymmetry by collecting the defender's information in advance. In an ideal case, the attacker concentrates on attacking those nodes with a higher benefit, e.g., the Leader node in the consensus mechanism, the Endorser, and Executor node with essential contracts and Anchor nodes in organizations. Multiple attack strategies are generally available for attackers to deal with different types of nodes at application layer. We formulate threats targeting at different types of nodes in consortium blockchain.

Threats to Executor Nodes: Assume that attackers can launch legitimate attacks by sending a large volume of invalid transactions at a small cost to block the node's transaction pool and deny the normal service of transaction execution. Attack benefit generally positively correlates with the number of contracts and logical nodes employed at the target node. Nodes will consider invalid transactions as such and exclude them from blocks, yet each node will still verify the validity of all received transactions. Empty blocks, adhering to the basic blockchain protocol rules, can be successfully added to the chain. Both invalid transactions and empty blocks are special transactions initiated by attackers within the consortium blockchain and cannot be restricted by built-in rate limits.

Threats to Endorser Nodes: the Timeout-manipulation attack [10], [26] mainly targets the Leader node and a specific number of Follower nodes. Attacking Leader node delays processing transaction messages sent by normal clients. Attacking a specific number of Follower nodes (at least $F+1$ nodes need to be attacked in the *Practical Byzantine Fault Tolerance* (PBFT) consensus) destroy the Leader replacement of the consensus system. There are not enough Follower nodes to vote for replacing the current Leader node.

B. Problem Definition

Consortium blockchain is only open to authorized organizations or institutions, and the joining and exiting of nodes usually require specific conditions defined by the governance rules, so that the total size of nodes in the consortium blockchain is smaller than that of the public blockchain. And the internal identity in the consortium blockchain is unforgeable, which creates a barrier for becoming an attacker. For example, Hyperledger Fabric has implemented a series of built-in rate restrictions to mitigate DDoS attacks. However, when attackers within the consortium blockchain control clients to initiate a large number of invalid transactions, such as empty blocks, normal generation of empty blocks is not hindered by built-in rate limitations.

The primary issue of this work is to prevent each node in organizations from attacks and lead to the minimum attack return, while reaching a Nash equilibrium between the attacker and organization. Rational attackers adopt an optimal DDoS attack strategy at each organization by selecting nodes to gain the maximum return at the instant state. Rational attackers can also take advantage of the feature of blockchain networks that allow anyone to freely send transactions and can launch legitimate

attacks by sending invalid transactions at a small cost. These invalid transactions may have meaningful purposes, such as simulating sending transactions to their own accounts when developing and testing blockchain applications to verify the format and logic of transactions, and sending transactions with zero transaction amounts may help ensure that accounts are not considered inactive or cleared. In addition, attack targets will be continuously changed after each round of MTD moves. Defenders learn MTD moves from DQN-based model and change states of nodes in each organization. Logically, the effect of MTD will be enhanced along with the increment of attack rounds. Our work has proved that there exists a Nash equilibrium for the situation above.

A necessary condition of terminating DDoS to an organization is all nodes in organization are survived from the attack. We define the problem as when the Nash equilibrium is reached, attacker's profit is as small as possible under the condition when all nodes in organization are not attacked.

We present the problem formally in Definition II.1.

Definition II.1: Minimizing Attack Profit Under Node-Saving Condition Problem (MAPOC): Inputs: include a set of attack profits for a node $\{a_{ij}^{be}\}$, a set of attack costs node $\{a_{ij}^{cost}\}$, a set contains the number of container deployed in each node $\{N_{ij}^{cont}\}$, the number of logical nodes deployed in each server $\{N_{ij}^{lon}\}$, a set contains the number of IP moving time for each node $\{M_{ij}^{IP}\}$, a set contains the number of port moving time in each node $\{M_{ij}^{Port}\}$ and a set contains initial malicious traffics interception rates of each node $\{\alpha_{ij}\}$ in each organization.

Outputs: include a set contains node attack profits $\{a_{ij}^{pro}\}$ and a set contains malicious traffics interception rates $\{\alpha'_{ij}\}$ when the Nash equilibrium is reached.

The proposed MAPOC problem is to find out optimal MTD actions to minimize the total attack profit (P_{total}) under the condition when none node in organization is attacked ($\alpha'_{ij} > \alpha_{ij}$).

Mathematical expression of MAPOC is given by (1).

$$P_{total} = \min \left[\sum_{i=1}^m \sum_{j=1}^n a_{ij}^{pro} \right] \quad \text{and} \quad \alpha_{ij} \geq \alpha'_{ij} \quad (1)$$

The subscript ranges in (1) are $i \in [1, m]$ and $j \in [1, n]$. $\min[\cdot]$ is the function that obtains the minimum attack profit value in (1). Regarding notations in an organization, we take the organization $\mathcal{O}_1$ as an example. The attack profit, benefit, and cost of the organization $\mathcal{O}_1$ are denoted by $P_1 = \sum_{j=1}^n a_{1j}^{pro}$, $j \in [1, n]$, $B_1 = \sum_{j=1}^n a_{1j}^{be}$, $j \in [1, n]$, and $C_1 = \sum_{j=1}^n a_{1j}^{cost}$, $j \in [1, n]$. $\mathcal{O}_i = \{\mathcal{O}_1, \mathcal{O}_2, \dots, \mathcal{O}_m\}$ denotes the set of organizations. The number of nodes in each organization is denoted by n . Value of profit a_{ij}^{pro} is obtained from $a_{ij}^{be} - a_{ij}^{cost}$. The profit of each organization is denoted in (2). When profit is less than 0, the DDoS attack will be ended for that organization.

$$P_i = B_i - C_i, \forall i \in [1, m] \quad (2)$$

For the benefit and cost of attacking nodes in MAPOC, we define the node attack benefit as a sum of the number of deployed containers (denoted as N^{cont}) and the number of running logical nodes (denoted as N^{lon}). Therefore, there is $a_{ij}^{be} = N_{ij}^{cont} + N_{ij}^{lon}$. MTD action changes victims' attack

surface, which increases attack cost. We define the node attack cost as a sum of the times of IP MTD action (denoted as M^{IP}) and Port MTD action (denoted as M^{Port}). Thus, there is a $a_{ij}^{cost} = M_{ij}^{IP} + M_{ij}^{Port}$.

Our model defines a malicious traffic interception rate α for traffic detection and filtering. The effective attack's cost is denoted by $a_{ij}^{cost}(1 - \alpha_{ij})$. The DDoS attacker needs to cost additional resources when malicious traffics are filtered. Assuming that k rounds of attacks are filtered, there exists a cost of attack to node p_{ij} is $a_{ij}^{cost}(1 + k\alpha_{ij})$. DDoS cost in p_{ij} needs to satisfy requirement of (3). Equation (4) defines α'_{ij} .

$$a_{ij}^{cost}(1 + k\alpha_{ij}) < a_{ij}^{be} \quad 1 \leq i \leq m, 1 \leq j \leq n \quad (3)$$

$$\alpha'_{ij} = \frac{a_{ij}^{be} - a_{ij}^{cost}}{ka_{ij}^{cost}} \quad 1 \leq i \leq m, 1 \leq j \leq n. \quad (4)$$

The proposed multistage game between DDoS attackers and defenders is explained in the followings.

Multistage Game Model between DDoS Attacker & Multiple Defenders. In our game model, players involve both organizations and attackers. The action space of each organization is the collection of MTD actions at each node from relevant organizations (refer to Section IV).

We use malicious traffics to measure the efficiency of DDoS defense, by taking a single organization $\mathcal{O}_k$ as an analysis object.

Considering the discount factor γ ($\gamma \in (0, 1)$), the accumulated reward of $\mathcal{O}_k$ is denoted by $V_k(s)$. $V_k(s)$ is essentially an expected reward function on player states, which is a function of state s . $V_k(s)$ is expressed in (5).

$$V_k(s) = E_\pi [P_k^s | s_t = s] \quad (5)$$

s_t is the state of P_k in MTD round t . The π denotes our DQN-based MTD strategy and the P_k^s is the reward of $\mathcal{O}_k$ in state s . Another representation of (5) is expressed by *Bellman Equation* in (6) (i.e., in a recursion format), in which the s'_k is the next state after the current action. The $r_{k,t}$ is the action reward of $\mathcal{O}_k$ in MTD round t and the s' is the next state of $\mathcal{O}_k$.

$$V_k(s) = r_{k,t} + \gamma V_k(s') \quad (6)$$

Similar to defenders, the reward of DDoS attackers is defined by (7), where $r_{a,t}$ and $V_a(s')$ denote bots' reward after MTD round t and accumulated reward in next state, respectively.

$$V_a(s) = r_{a,t} + \gamma V_a(s'). \quad (7)$$

The operating principle of our gaming model is formulating multi-phase interactions between attackers and organization defenders in a consortium blockchain. Organizations in the same channel play as defenders, each of which adopts a specific MTD action (obtained from Reinforce learning) based on the attacker's strategy. The DDoS attacker changes DDoS targets according to defenders' states. We assume the attacker takes an optimal attack strategy at every step to maximize the attack benefit.

The proposed game model is essentially a limited Non-cooperative mixed strategy game. According to *Nash Equilibrium* [27], the proposed game model has at least one Nash equilibrium point. Specifically, the Nash equilibrium point can be expressed as the states of every defender and the DDoS

attacker. Suppose there exists a time spot t^* that arrives the Nash equilibrium, state of defender $\mathcal{O}_k$ is s_k^* , which is the states collection of all internal nodes. Equation (8) shows the occasion (t^*) when the Nash equilibrium is reached. Nash equilibrium point can be expressed by $(\{s_1^*, s_2^*, s_3^* \dots s_m^*\}, s_a^*)$.

$$\begin{cases} V(s_k^*) \geq V(s_k) & \text{The defender } \mathcal{O}_k \\ V(s_a^*) \geq V(s_a) & \text{The DDoS attacker} \end{cases} \quad (8)$$

Next, we prove the existence of Nash equilibrium for the proposed model by proving presence of potential game [28]. The key is to prove the existence of function called $\phi(a)$, which satisfies (9).

$$V_{a-i}(a_i) - V_{a-i}(a'_i) < 0 \Rightarrow \phi_{a-i}(a_i) - \phi_{a-i}(a'_i) < 0 \quad (9)$$

In (9), $V_{a-i}(a_i)$ denotes the sum of accumulated benefit except the i -th player (when i -th player with an action a_i). The $i \in [1, k]$ denotes the defenders; otherwise, the player is the attacker. The a'_i is another action, which satisfy (11).

$$r_{a-i}(s, a_i) < r_{a-i}(s, a'_i) \quad (10)$$

We further define the potential function of the proposed game in (11).

$$\begin{aligned} \phi_{a-i}(a_i) = & \sum_{k=1, k \neq i}^m [r(s, a_i) + \gamma V(s', \pi_k)] I_{i \in [1, k]} \\ & + [r(s_a, a_i) + \gamma V(s'_a, \pi_a)] I_{i \notin [1, k]}. \end{aligned} \quad (11)$$

An I is the indicator function that is valued 1 when the condition is satisfied and is valued 0 if not. We formulate the assumption by Theorem II.1.

Theorem II.1: The proposed multistage game between multiple defenders and the DDoS attacker is a potential game with the potential function of (11).

Proof: We prove this theorem on both defenders and attackers, including $i \in [1, k]$ (defenders) and $i \notin [1, k]$ (attacker). First, for the DDoS defenders ($i \in [1, k]$), the potential function is shown in (12).

$$\phi_{a-i}(a_i) = \sum_{k=1, k \neq i}^m [r(s, a_i) + \gamma V(s', \pi_k)] \quad (12)$$

The operation $\phi_{a-i}(a_i) - \phi_{a-i}(a'_i)$ is expressed by (13).

$$\begin{aligned} \phi_{a-i}(a_i) - \phi_{a-i}(a'_i) &= \sum_{k=1, k \neq i}^m [r(s, a_i) + \gamma V(s', \pi_k)] \\ &\quad - \sum_{k=1, k \neq i}^m [r(s, a'_i) + \gamma V(s', \pi_k)] \\ &= \sum_{k=1, k \neq i}^m [r(s, a_i) - r(s, a'_i)] = r_{a-i}(s, a_i) - r_{a-i}(s, a'_i) < 0 \end{aligned} \quad (13)$$

Therefore, Theorem II.1 is proved in the condition of $i \in [1, k]$. For the DDoS attacker ($i \notin [1, k]$), the potential function is depicted in (14).

$$\phi_{a-i}(a_i) = [r(s_a, a_i) + \gamma V(s'_a, \pi_a)] I_{i \notin [1, k]} \quad (14)$$

For different actions of DDoS attackers, the states of defenders are constant in the assumption. The consequence of the computation $\phi_{a-i}(a_i) - \phi_{a-i}(a'_i)$ is shown in (15).

$$\begin{aligned}\phi_{a-i}(a_i) - \phi_{a-i}(a'_i) &= r(s_a, a_i) - r(s_a, a'_i) \\ &= r_{a-i}(s_a, a_i) - r_{a-i}(s_a, a'_i) < 0.\end{aligned}\quad (15)$$

Therefore, Theorem II.1 is proved when $i \notin [1, k]$. $\square$

C. Overview

In our approach, when an organization detects a DDoS attack, victims will initiate transactions to synchronize attack data between organizations via smart contracts. To prevent malicious internal clients from uploading error information to disrupt the defense, we set up a special proxy node in each organization to synchronize DDoS data, which is only called by authenticated participant organizations and cannot be discovered by other internal clients. When an organization detects DDoS attacks, the proxy node in the victim organization will synchronize the obtained attacks information/data with other organizations by initiating transactions, such as bots IPs, attack types, attack intensity, and attack duration. The traffics sent from bots can be successfully filtered once other organizations update the strategy. Moreover, the persistent malicious traffic generally cannot be directly detected or filtered, as the attacker continuously adjust the attack resource following the constantly changing environment of the defender in order to obtain better returns. In order to address this type of attack, we utilize the Deep Q-network (DQN) to generate an MTD strategy in each organization. We represent the DDoS malicious traffics by attack type, attack intensity, and attack duration. A variety of MTD hopping methods are covered in our approach, such as hopping, port hopping, container hopping, and logic node hopping. With the increment of happenings of malicious traffics, our approach will continuously update its filter strategy to improve the defense accuracy. Finally, we adopt a multi-round optimized MTD strategy to defend persistent malicious traffics.

A fundamental philosophy of our approach is to formulate the attack-defense interactions between the attacker and defender. We describe the interaction via formulating a multistage game model [29]. Once a DDoS attack occurs, the nodes of defenders first filter out detected malicious traffics, retrieving the knowledge from the ledger. When persistent malicious traffics attack occurs, the MTD method that follows the local DQN training result can be adopted to mitigate the threats. The DQN is trained by malicious traffics collected by all organizations. In this case (after MTD actions), attackers discover the new environment of each node in consortium blockchain, so that the attack strategy at each node will be adjusted for achieving an optimal attack return. In the meantime, defenders' nodes synchronize the new updated attack data to the ledger via implementing a smart contract. The interactions above are repeated to increase the defense performance. Therefore, the interaction between DDoS attackers and defenders are formulated into a multistage game, such that a Nash equilibrium eventually can be reached. After the repeated executions of MTD, the rational attacker's cost will increase. Logically, when the attacker's gain from the defender's node becomes negative, the attack will be terminated.

III. CONTRACT-BASED COLLABORATIVE DEFENSE

We define a special smart contract that can only be called by internal nodes in organizations. Transactions with attacker information or attack data are generated by smart contract. The DDoS attack information will be recorded by each organization when the consensus and verification are made.

A. Architecture

In a consortium blockchain, only necessary business information can be synchronized between organizations due to the autonomy and independence of each organization. Therefore, information on bots and malicious traffics cannot be directly synced across organizations, which makes the information asymmetry between internal DDoS bots and victimized nodes in consortium blockchain.

Malicious traffics from internal bots are essentially as the same as normal traffic. The typical SDN-based programmable switch defenses and firewalls cannot completely filter malicious traffics. Typical Internet DDoS defenses cannot defend DDoS attacks launched by nodes inside different organizations bypassing the identity authentication of the consortium blockchain. When the scale and intensity of attacks launched by internal attackers have not yet reached the Internet DDoS defense standards, insider attackers can successfully slow down the transaction processing and block generation services of the consortium blockchain. Attackers do not need to launch a full DDoS attack at a single node, which is much easier than an Internet DDoS attack. The incremental amount of the attacked nodes will further negatively influence the whole efficiency of the blockchain system due to a slower speed of block generations caused by DDoS attacks. Thus, traditional Internet DDoS defense is inapplicable for the proposed attack scenario.

We proposed a smart contract-based DDoS attack data sharing and collaborative defense method, the first defense method in the proposed defense approach. Internal DDoS attackers not only launch typical DDoS attacks, but also control internal clients to collect information from defenders in advance to formulate the best attack strategy, making malicious traffic ignore current identity verification, threaten transaction execution, consensus, and prevent submissions, thereby achieving maximum attack benefits. We have defined a special smart contract to synchronize DDoS attack information that is only called by internal blockchain nodes. The proposed contract generates the transactions with attackers or malicious traffic information. Each organization saves DDoS attack information after the consensus and verification, which is as same as normal transactions.

The contract-based collaborative method between organizations takes advantage of quickly committing blocks by smart contract in consortium blockchain. However, the problem of the denial of service for synchronizing attack data should be solved. In response to the threat of compromised malicious internal nodes uploading wrong bot information to disrupt the defense, we set a special proxy node in each organization for synchronizing DDoS data, including the transactions executing, consensus, and committing blocks, which can be only called between organizations and cannot be found by internal clients. The proxy node also broadcasts the blocks with DDoS data to

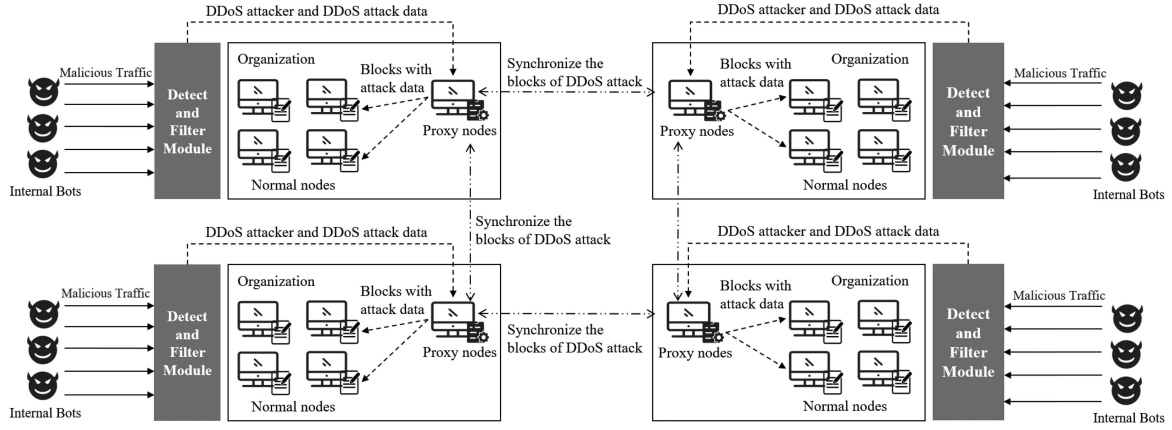


Fig. 2. Architecture of contract-based collaborative defense method.

other nodes in the same organization. The proxy node ensures there is at least one ledger that keeps complete attack data in each organization, which ensures the integrity of attack data [30], [31] in the whole consortium blockchain. The collaborative defense method is shown in Fig. 2.

Specifically, each organization's traffic detection and filter module calls the function of `UPLOAD_ATTACK` in contract of DDoS recording, which uploads the DDoS attack data by transactions. The proxy node broadcasts the launched transactions to other proxy nodes in organizations. These transactions will be blocked and broadcast to other organizations for synchronizing the attack data and information of attackers.

B. Functionality

The smart contract for DDoS attacks records the malicious traffic information. The bots have specific node names ($\mathcal{N}^b$) and public keys (P_k^b) since the bots are authenticated clients. In addition, the attacker's address ($\mathcal{A}^b$) and state (malicious or normal, denoted by Sta^b) are also recorded in the distributed ledger. The bot can be represented by $bot = \langle \mathcal{N}^b, \mathcal{A}^b, P_k^b, Sta^b \rangle$. A tuple with five parameters are used to record the DDoS attacks data, which is shown as $\langle \mathcal{A}^b, \mathcal{T}^a, \mathcal{I}^a, p_{ij} \rangle$. In the attacks data tuple, the parameters represent the address of the bot, the DDoS attack method, the attack intensity and the victim's ID.

The operation of the proposed smart contract includes uploading attack data, obtaining node information and changing the state of internal clients, all of them are invoked by the traffic detection and filtering module. Three operations are described as follows.

- **UPLOAD_ATTACK**: Uploading the detected attack data to the ledger and synchronizing the data to other organizations. The form of attack data is a tuple with five parameters.
- **GET_CLI_INFO**: Defenders obtain the client's state from the contract when detecting the traffic. The traffic from malicious clients will be filtered directly.
- **CHANGE_NODE_STATE**: This function sets the node's state to malicious if the node is detected as malicious (sending malicious traffic). In addition, this function also sets the node's state to normal if the node's traffic return to normal.

```
func MAIN_UPLOAD_ATTACK( AttackInfo
    attackInfo ) {
    transaction := &Transaction{
        AttackInfo: attackInfo ,
    }
    UploadDDoSInfo( transaction )
    return
}
// Uploading to the ledger by contract.
func UploadDDoSInfo( Transaction
    transaction ) bool {
    res := Contract.UPLOAD_ATTACK(
        transaction )
    return res
}
```

The following is a detailed presentation of contract execution and code snippets.

UPLOAD_ATTACK: First, the traffic detection and filtering module sends transactions with detected malicious traffic and malicious attacker to proxy node. Attacker information is presented as $\langle \mathcal{N}^b, \mathcal{A}^b, P_k^b, Sta^b \rangle$ and the DDoS attack data is presented as $\langle \mathcal{A}^b, \mathcal{T}^a, \mathcal{I}^a, p_{ij} \rangle$. Second, the proxy node executes the transactions and generates the blocks. The transactions and corresponding results will be sent to proxy nodes in other organizations. The proxy node packages the received transactions into blocks and broadcasts them to other proxy nodes. In the last, proxy nodes verify the received blocks and decide whether to save the block or not. The core logic of `UPLOAD_ATTACK` is shown in code `MAIN_UPLOAD_ATTACK`.

In the code above, the function of `DetectionAndCollect` is used to detect and collect malicious traffics, which is not essential in this paper. The function of `UploadDDoSInfo` calls the function `UPLOAD_ATTACK` in the contract, which commits the transactions to proxy for the next processing. The `UploadDDoSInfo` returns true if the transaction is successfully saved by blockchain.

GET_CLI_INFO: This function is called before traffic detecting and filtering. The traffic detection and filter module obtains

the DDoS data from contract first and checks whether there are bots in current traffic. The current malicious traffics from the bots will be filtered directly. The core logic of GET_CLI_INFO is shown in code *MAIN_GET_CLI_INFO*.

```
func MAIN_GET_CLI_INFO() {
    // Getting saved bots information
    bots := GetBotsInfo()
    // Traversing the current clients
    for _, client := range
        CurrentClientsCollection {
        // Checking whether the client
        // is bot or not
        if CheckIsBots(client, bots) {
            Filtering(client)
        }
    }
    return
}
// Getting saved bots information by
// contract
func GetBotsInfo() []Bots {
    bots := Contract.GET_CLI_INFO()
    return bots
}
```

The traffic detection and filtering module call function of GetBotsInfo to obtain the saved bots and calls function of CheckIsBots to check whether there are bots in the current client node.

CHANGE_NODE_STATE: The traffic detection and filtering module dynamically changes the state (normal or malicious) of internal clients according to whether current traffic is malicious. The core logic of *CHANGE_NODE_STATE* is shown in code *MAIN_CHANGE_NODE_STATE*.

```
func MAIN_CHANGE_NODE_STATE() {
    clients, states := DetectStateChange()
    transaction := &Transaction{
        Client: clients,
        State: states,
    }
    res := ChangeCliState(transaction)
}
// Modifying the state of internal.
func ChangeCliState(Transaction
    transaction) bool {
    res := Contract.CHANGE_NODE_STATE(
        transaction)
    return res
}
```

The contract caller generates a transaction based on the client's state (by function DetectStateChange) and calls

the function ChangeCliState to launch the transaction. ChangeCliState returns a *True* when the transaction is successfully executed.

IV. DQN-BASED MTD STRATEGY GENERATION

A. Method

Smart contracts collect and store malicious traffic data within each organization. The effectiveness and accuracy of this approach will increase as more malicious traffic is identified. Once an internal attacker gains access, it may use external botnets to control nodes and launch DDoS attacks. Attack patterns are varied in type, intensity, or duration, depending on the node type. Since attacker identities within the consortium blockchain cannot be forged, we must discretize the state space to prevent morphing through stealth strategies, e.g., using different Sybil accounts and IPs. Each organization uses a training model that incorporates malicious traffic detected by others, but direct malicious traffic carries greater weight. Our primary concern is internal nodes within the consortium blockchain, as they can control internal clients to gather information from defenders, formulate optimal attack strategies, and bypass identity verification. This threatens transaction execution, consensus, and prevents submissions, maximizing attack benefits. Thus, attacks must evade both traditional DDoS defenses and our dynamic defense strategies to succeed. Rational attackers will cease their attacks when the costs outweigh the benefits.

1) *State Space*: We focus on the blockchain node p_{ij} in organization Org_i , the DDoS attacker pays attention to the types, the attacking cost and profit in time slot t , which can be expressed as $p_{ij}^t = \langle p^{Type}, \{bot\}^{cost}, a^{pro} \rangle_{ij}^t$. The p^{Type} is a set of types because one node may display multiple roles of consortium blockchain; for example, $p_{ij}^{Type} = \langle Endorser, Committer \rangle$ means the p_{ij} provides the transaction execution service and blocks storage service. The malicious traffics in time t in p_{ij} called r_{ij}^t can be denoted by $r_{ij}^t = \{ \langle A^b, T^a, I^a, D^a \rangle \}_{ij}^t$, which is the collection of all malicious traffics to the p_{ij} . The state of p_{ij} in time t denoted by the $S_{ij}^t = \langle p_{ij}^t, r_{ij}^t \rangle$, which is the node state in the analysis of DQN learning.

2) *Action Space*: According to the MTD theory and the features of consortium blockchain, the attack surface of blockchain node can be expressed as two parts, including the service addresses and the logic nodes deployed in the physical node. The service addresses of node p_{ij} in time t are denoted by $M_{ij}^t = \langle M^{IP}, M^{Port}, M^{Cont} \rangle_{ij}^t$. M^{IP} denotes the exposed IP address for providing blockchain service. M^{Port} is the set of ports for multiple types of blockchain service, and the M^{Cont} is the set of containers for running multiple smart contracts. The multiple logic nodes can be deployed in one physical node, denoted by L_{ij}^t . We use AS_{ij}^t to denote the attack surface of p_{ij} in time t , where the $AS_{ij}^t = \langle M_{ij}^t, L_{ij}^t \rangle$.

The action in proposed DQN-based MTD is the change of AS_{ij}^t as the time sequentially, including the change of M_{ij}^t , L_{ij}^t or both of them. The state is changed by defense strategy as time grows according to DQN results, the state change from t to Δt can be denoted as $\langle M_{ij}^t, L_{ij}^t \rangle \Rightarrow \langle M_{ij}^{t+\Delta t}, L_{ij}^{t+\Delta t} \rangle$. The specific

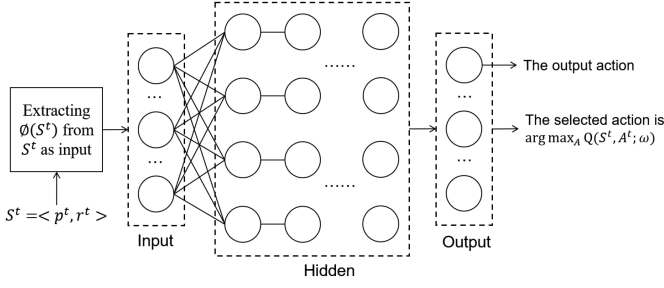


Fig. 3. DNN workflow for training.

transition process is shown in (16).

$$\begin{cases} M_{ij}^t \neq M_{ij}^{t+\Delta t}, L_{ij}^t = L_{ij}^{t+\Delta t} & M_{ij} \text{ moved, } L_{ij} \text{ unchanged} \\ M_{ij}^t = M_{ij}^{t+\Delta t}, L_{ij}^t \neq L_{ij}^{t+\Delta t} & M_{ij} \text{ unchanged, } L_{ij} \text{ moved} \\ M_{ij}^t \neq M_{ij}^{t+\Delta t}, L_{ij}^t \neq L_{ij}^{t+\Delta t} & M_{ij} \text{ moved, } L_{ij} \text{ moved} \end{cases} \quad (16)$$

3) *Reward Function*: We evaluate the DDoS effect in node p_{ij} by Maximum Likelihood Estimation (MLE), the rate arriving obeys the Poisson distribution, and the queue model is denoted as M/M/1. The probability density function of the Poisson distribution is defined as (17).

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, k \in \mathbb{N} \quad (17)$$

The λ denotes the number of malicious packets arriving in the unit time. The expectation of Poisson distribution is λ , which can be used as the evaluation of DDoS effect in each node. The λ_{ij} denotes the DDoS attack strength to p_{ij} .

According to MLE, the sample of the last β time units about malicious traffics are denoted by $\{X_1, X_2, \dots, X_\beta\}$, the likelihood function is shown in (18).

$$L(\lambda|X_1, X_2, \dots, X_\beta) = f_\lambda(X_1, X_2, \dots, X_\beta) \quad (18)$$

The value of λ that makes the maximum f_λ is the target value for estimating the traffic in node p_{ij} . The f_λ is also denoted in probability format, which is shown in (19).

$$f_\lambda(X_1, X_2, \dots, X_\beta) = P(X_1, X_2, \dots, X_\beta|\lambda) \quad (19)$$

Defense reward of defender O_k is the expected value of all nodes in O_k , which is shown in (20).

$$R_k = \frac{1}{n} \sum_{j=1}^n \frac{1}{\lambda_{kj}}. \quad (20)$$

B. Workflow of Deep-Q-Network Learning

The DNN is used for performing the Deep-Q-Network Learning process, which generates the MTD method for the specified blockchain node in organization. The interaction between the DQN agent and environment is shown in Fig. 4. DQN agent includes Target Q-network and Main Q-network, the Loss Function is used for approaching the optimal action. The environment includes malicious traffics monitoring and the current logic nodes of p_{ij}^t . The reward of changing state to S_{ij}^{t+1}

comes from the current environment evolving with persistent malicious traffic.

The agent does not have action experience in the first epoch, but the malicious traffics are distributed to attack different nodes. One physical consortium blockchain node may play multiple roles at the beginning of DDoS attack, the MTD action could be selected randomly because there is no experience with malicious traffics. After the first action is taken, the environment can obtain the action reward. Finally, the reward and the new state S_{ij}^{t+1} are sent simultaneously to Main Q-network and the Replay memory D. With the continuous changes in the defense environment, the persistent malicious attacks aimed at maximizing profits are also constantly adjusting their attack strategies, achieving a Nash equilibrium between defense and attack strategies. When the attack cost exceeds the attack benefit, the multi-round optimized MTD strategy can successfully counter persistent malicious traffic.

The DNN architecture is used to train the state information and generate the value of every output action in epoch t . The actor-network needs to take the exploration for obtaining the next state by the probability $p_\epsilon = \epsilon$. The experience replay and target network are also applied to make learning process more stable. The experience replay is called D for storing the history state transition, which will be extracted as a minibatch to train our MTD DNN. Each transition can be expressed by $\langle A^t, R^t, S^t, S^{t+1} \rangle$ in transition set. The correlation between the continuous malicious attack samples is reduced, alleviating the influence of continuous DDoS attack. The target network is copied by main Q-network each C steps, which reduces the loss value and approaches the optimal solution.

Fig. 3 shows the DNN workflow for training. The state parameters need to be extracted as the DNN input. The parameters of p_{ij}^t and r_{ij}^t are concatenated as the format for DNN processing. We used the ReLU as the activation function for building the hidden layers. The outputs of DNN are the Q value of each state-changing action, which is also the probability of subsequent action. In addition, the following action with the most Q value will be selected and taken in the related blockchain node, like changing the endorsement port of peer node p_{ij} . The change of p_{ij} attack surface influences the node state, the cost of attacking to the p_{ij} endorsement service is increased and the DDoS attack type is changed due to the port changing. In the case where internal DDoS attackers disrupt the normal operation of the consortium blockchain driven by external interests, the attack will terminate when the cost of launching the attack and the benefits obtained are unequal or even less.

Detailed training workflow is shown in Algorithm 1, which is the process of one learning episode. There are a number of MAX_STEPS in each episode and the episode ends if $IsEnd$ is True. The loss function is shown in (21).

$$L(\omega) = \frac{1}{m} \sum_{k=1}^m (y_k - Q^*(\phi(S'_k), A'_k, \omega))^2 \quad (21)$$

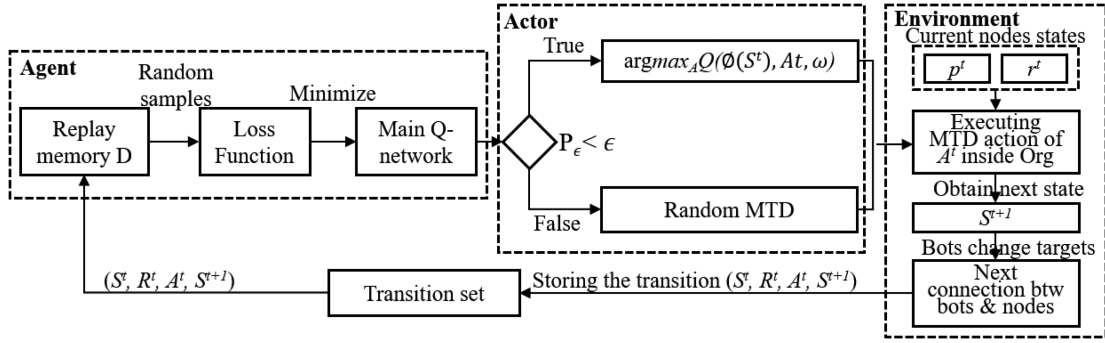


Fig. 4. The DQN workflow with the blockchain node states.

TABLE I
DDoS ATTACK PARAMETERS

Payload type	Methods	Intensity (pps)
TCP Flood	SYN Flood	10000, 20000, 30000, 40000, 50000
UDP Flood	Amplification	1000, 2000, 4000, 8000, 10000
ICMP Flood	Ping Flood	1000, 2000, 4000, 8000, 10000

There are m number of samples randomly selected from Replay memory D. The y_k is denoted in (22).

$$y_k = \begin{cases} R_k & \text{If terminates in next step} \\ R_k + \gamma \max_{A'} Q^*(\phi(S'_k), A'_k, \omega^*) & \text{Otherwise.} \end{cases} \quad (22)$$

The ω and ω^* denote the parameters in DNN network. Specifically, the ω^* denotes the target network parameter, which will be updated to decrease the loss function value.

V. EVALUATION AND THE RESULTS

A. Experiment Configuration

Evaluations were carried out under Python 3.10.4 version and PyTorch 1.10.2 version. Simulation environment included an Intel Core i5-7200 U CPU, a Windows 11 operation system with an 8.0 GB main memory. We compared our approach with Bayesian Q-Learning Game Approach [32], MOTAG [33], DSEOM mechanism [15], and the theoretically optimal results (Up Bound). Our experiment used 100 nodes, distributed in 10 different organizations, to communicate over the Internet to simulate the actual situation of cross-regional operations. Each node is equipped with a 4-core CPU, 4 GB of memory, and 100 GB of SSD storage. Experiment settings were summarized in Table II. Evaluations of single organization and multiple organizations were separately processed. All of these evaluations are conducted on the Hyperledger Fabric v2.4.0 platform.

In addition, we used BoNeSi to generate flood attacks (including ICMP, UDP and TCP), the parameters such as attack intensity, bots scale, target address and attack duration can be configured through options. The parameters of the DDoS attacks are shown in Table I. We also adopted the HULK [34] attack as one of the DDoS attack methods, randomly generating forged parameter names and values.

TABLE II
EXPERIMENT SETTINGS (INS: INITIAL NODE STATES; SO: SINGLE ORGANIZATION; MO: MULTIPLE ORGANIZATIONS; NN: THE NUMBER OF NODES)

Settings	INS (S_{ij})	# Orgs (m)	# Org. nodes	# bots	Episodes
SO	#1.1 $\langle 1, 1, 100, 100 \rangle$	1	10	100	100
	#1.2 $\langle 1, 1, 100, 100 \rangle$	1	10	100	1000
	#1.3 $\langle 1, 1, 100, 100 \rangle$	1	10	500	1000
	#1.4 $\langle 1, 1, 100, 100 \rangle$	1	10	1000	1000
MO	#2.1 $\langle 1, 1, 100, 100 \rangle$	2	10	400	1200
	#2.2 $\langle 1, 1, 100, 100 \rangle$	4	10	800	1200
	#2.3 $\langle 1, 1, 100, 100 \rangle$	6	10	1200	1200
	#2.4 $\langle 1, 1, 100, 100 \rangle$	8	10	1600	1200
	#2.5 $\langle 1, 1, 100, 100 \rangle$	10	10	2000	1200
NN	#3.1 $\langle 1, 1, 100, 100 \rangle$	20	2	1000	1200
	#3.2 $\langle 1, 1, 100, 100 \rangle$	20	4	1000	1200
	#3.3 $\langle 1, 1, 100, 100 \rangle$	20	6	1000	1200
	#3.4 $\langle 1, 1, 100, 100 \rangle$	20	8	1000	1200
	#3.5 $\langle 1, 1, 100, 100 \rangle$	30	2	1000	1200
	#3.6 $\langle 1, 1, 100, 100 \rangle$	40	4	1000	1200
	#3.7 $\langle 1, 1, 100, 100 \rangle$	50	6	1000	1200
	#3.8 $\langle 1, 1, 100, 100 \rangle$	60	8	1000	1200

Each attacker will choose one of the payload types and methods in attack round, and the attack intensity will be limited in corresponding range. We set the attack duration to 60 seconds.

1) *Internet DDoS Defense*: To evaluate whether the Internet DDoS defense can successfully intercept malicious traffic, we monitored the device status of the target node under attack and set the average CPU utilization, network bandwidth, maximum TCP connections and attack time as parameters. We launched DDoS attacks from the beginning and initiated normal block transactions after the 14th second.

As shown in Fig. 5, the performance of the target device has not changed specifically before the normal transaction is launched, which indicates that the DDoS malicious traffic launched by the attacker will be intercepted by the Internet DDoS defense.

Meanwhile, in order to prove the practicability of DDoS attacks on blockchain nodes after DDoS protection on the Internet, we mainly evaluated two different DDoS attacks launched by attackers inside and outside the consortium blockchain. We set the transaction number and block time as parameters. In the two scenarios of internal and external DDoS attacks, the normal node initiates block transactions according to the transaction number, and the corresponding time of the block reflects the service

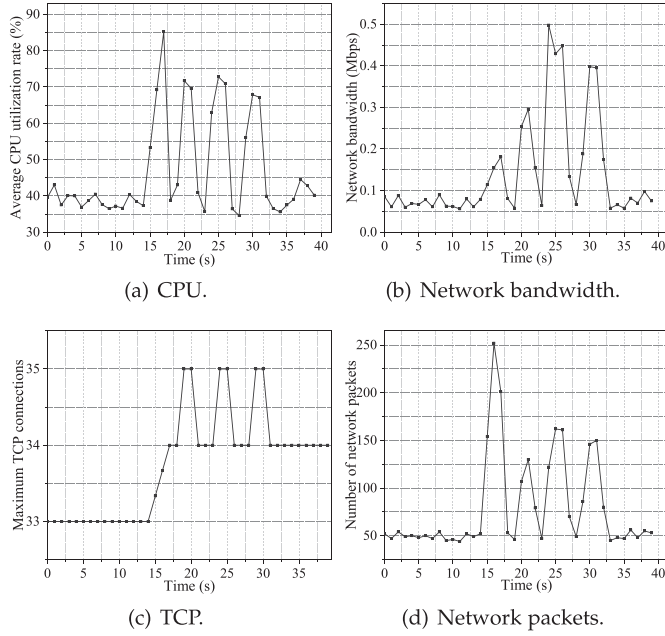


Fig. 5. Performance of the attacked target device.

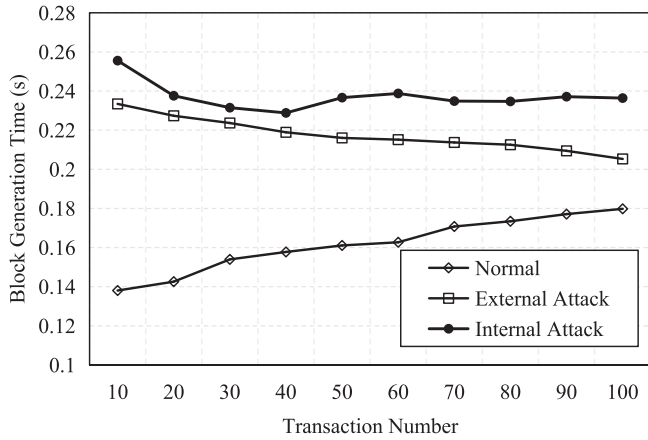


Fig. 6. Effects of different attack modes under Internet DDoS defense.

quality of the consortium blockchain, indirectly reflecting the defense effect of the DDoS protection method on the Internet against internal and external DDoS attacks.

After 100 transactions, the average block generation time for the three attack scenarios is 0.18 s, 0.21 s, and 0.24 s. The results of the Fig. 6 showed that DDoS attacks initiated by internal attackers have a greater negative impact on consortium blockchain than external attacks. DDoS attacks launched by external attackers have a weak impact on the consortium blockchain, and external malicious traffic will be identified and discarded by the Internet DDoS defense. However, DDoS attacks initiated by internal attackers can have a negative impact on consortium blockchain services. The main reason is that although the frequency of invalid transaction DDoS generated by internal attackers is lower than that of external attackers, the computing power resources of the consortium blockchain

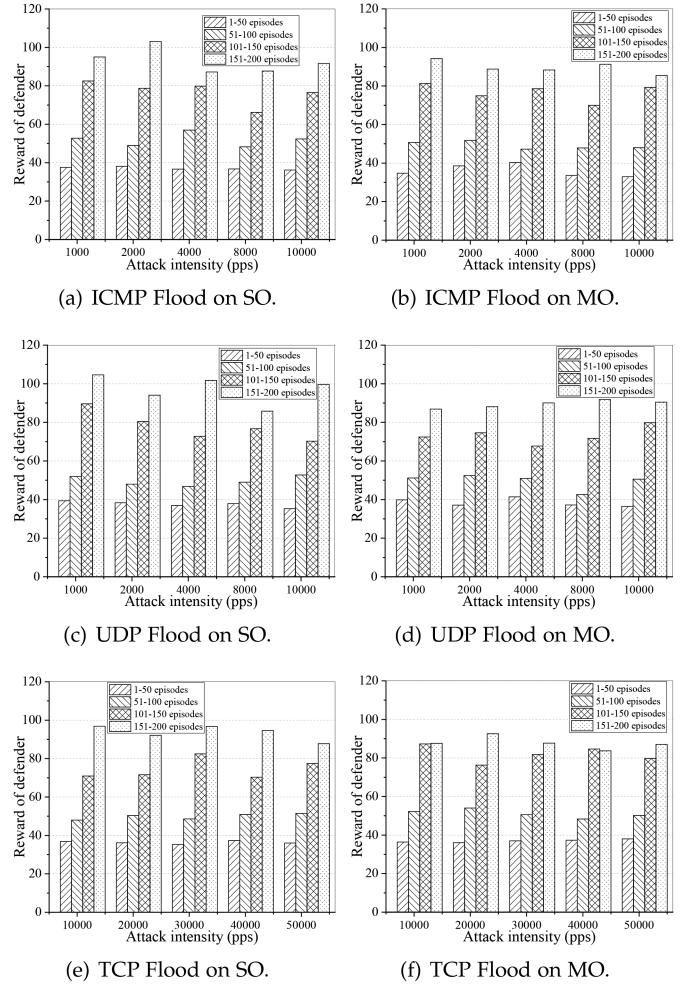


Fig. 7. The impact of different types of DDoS attacks on the rewards of Single Organization (SO) and Multiple organizations (MO) defenders.

nodes are limited, and the throughput is low. Continuously processing block transactions in a short period of time can cause consensus pressure, resulting in increased block generation time and decreased service quality. The experimental results show that Internet DDoS defense can properly deal with external attackers, but it is powerless to deal with DDoS attacks launched by internal attackers.

As shown in Fig. 7, regardless of whether in single organization or multiple organizations, the rewards of defender increased as episodes progress. It indicated that defenders are capable of effectively countering DDoS attacks of varying types, methods, and intensities launched by attackers.

2) *Single Organization*: In a single organization, we mainly evaluated the relationship between the attacker's and defender's benefits with DQN learning episode. We set the number of nodes and the number of bots as parameters. Rational DDoS attackers terminate attacks as long as reaching the requirement of (2). Equation (2) is an episode-stopping signal, which means an episode of DQN would stop only when reaching a DDoS termination condition.

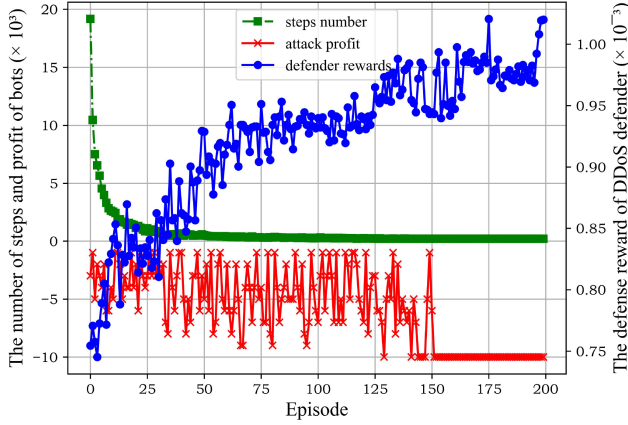


Fig. 8. Defender's rewards and attacker's profits inside one organization by each episode.

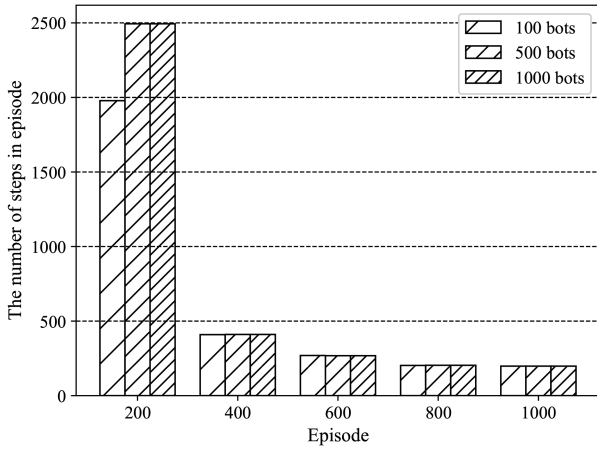


Fig. 9. Learning steps in each episode with the different number of bots (100, 500 and 1000).

There are four MTD attributes of blockchain node state, including IP moving times (M_j^{IP} , maximum value is 100), port moving times (M_j^{Port} , maximum value is 100), the number of developed containers (N_j^{cont} , minimum value is 1), and the number of logic nodes (N_j^{lon} , minimum value is 1). The initial state of nodes is shown in Setting #1.1. The profit of DDoS attacker is the difference between the benefit and attacking cost of the blockchain node; the benefit of attacked organization is the sum of benefits of attacked nodes.

The attacker adopted the optimal DDoS attack method, which selected nodes with the most benefit as targets in sequence (See Setting #1.1). Memory pool size was 200 and minibatch size was 16 (refer to Fig. 8).

We also evaluated the relationship between the number of bots and the DQN efficiency. Our evaluation reflected the defense efficiency in different scales of botnet; practicability of the proposed DQN-based defense could be reflected. We set the number of bots as a variable of the experiment; the experiment data was the same as the last experiment. The number of steps in each episode was evaluation metric in this evaluation. We carried out three times of experiments with different numbers of bots. The settings are given in Settings #1.2-#1.4, refer to Fig. 9.

Algorithm 1: The MTD Action Selection Method.

Require: The initialized replay memory D , the initialized main Q network and target Q network with ω and ω^* , the existing unhealthy states caused by DDoS.

Ensure: $\{S', \omega^*, \omega\}$

```

1: procedure Deep Q-learning Episode  $D, \omega, \omega^*, S$ 
2:   for  $step = 1 \rightarrow MAX\_STEPS$  do
3:     Initializing the first state  $S = \langle p^0, r^0 \rangle$  as  $\phi(S)$ .
4:      $IsEnd \leftarrow False$ 
5:     Input the  $\phi(S)$  to main Q-network and obtain the
       output actions, selecting the action  $A$  according to
        $\epsilon$ -greedy method.
6:     Execute action  $A$  (defined in Section IV-A3), the
       new state  $S'$ ,  $\phi(S')$  and reward  $R$  are obtained.
7:     if  $R \leq 0$  then
8:        $IsEnd \leftarrow True$ .
9:     end if
10:    Put the tuple  $\langle \phi(S), A, R, \phi(S'), IsEnd \rangle$  into
       replay memory  $D$ .
11:    Renew the current state for the node,  $S \leftarrow S'$ .
12:    The number of  $m$  samples are selected randomly
       from  $D \langle \phi(S_k), A_k, R_k, \phi(S'_k), IsEnd_k \rangle$ ,
       computing the target Q value of  $y_k$  according to
       (22).
13:    Computing and minimizing the loss function (21)
       and updating  $\omega$  in each step.
14:    Update the  $\omega^*$  by  $\omega \leftarrow \omega^*$  in each  $C$  steps.
15:    if  $step \bmod C == 0$  then
16:       $\omega \leftarrow \omega^*$ 
17:    end if
18:    if  $IsEnd == True$  then
19:      return
20:    end if
21:  end for
22: end procedure

```

Next, we evaluate the effect of sharing frequency of attack information is shared on defense reward. This experiment was carried out in a single organization. We set the sharing intervals are 100 ms, 500 ms, 1000 ms, 5000 ms, and 10000 ms respectively. The Settings#1.2 contains other parameters. The expectation of this evaluation is defender's reward will increase with the increment of sharing frequency of malicious traffics, but the defense reward will decrease after reaching a certain value. The specific results are shown in Fig. 15.

Finally, we conducted defense performance under different attack intensities. We used simulated malicious clients to generate malicious traffics with different attack intensity and recorded the number of saved nodes in each round. We conducted four experiments with different attack traffic rates, the traffic rates of each malicious client were 2000 pps, 3000 pps, 4000 pps and 5000 pps. Other configuration parameters are #1.4 in the Table II. The experimental results are shown in Fig. 14.

3) *Multiple Organizations*: DDoS defense efficiency in different numbers of organizations (m) was also researched. Based

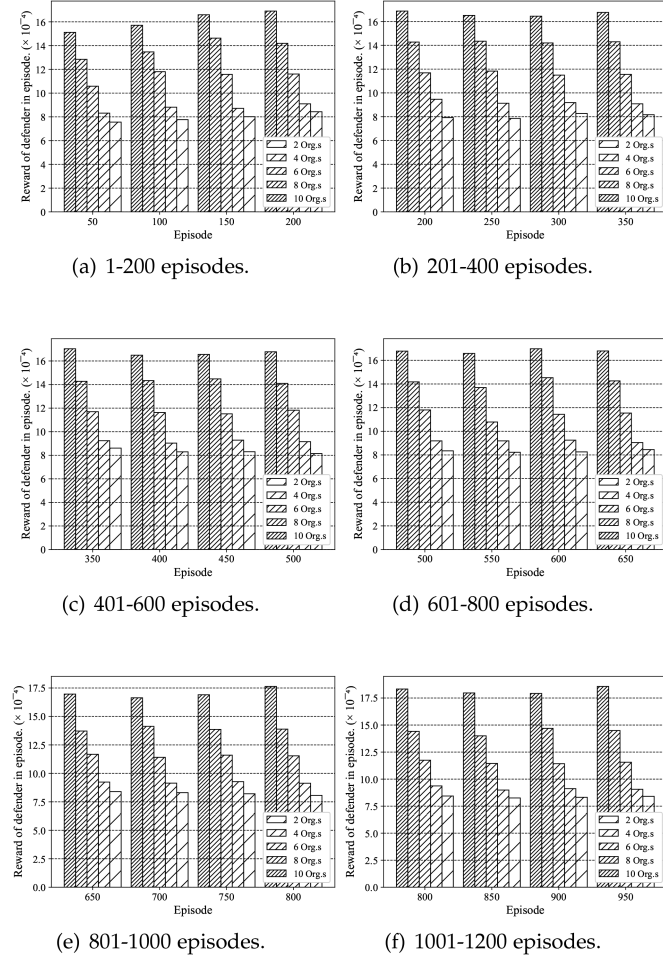


Fig. 10. Rewards of defender in episode with the different number of organization.

on the proposed theory, the defense mechanism in this paper can achieve better defense effects in the case of multiple organizations, because each defender can obtain more malicious traffics information. Therefore, defenders can directly filter out more malicious traffics and train a more accurate deep reinforcement learning model based on these traffic data. In addition, the defender can obtain various types of attack information, which can improve the elasticity and scalability of the defense model. The difference between the collaborative situation and the single organization situation was the scale of training data. The collaborative situation obtained more malicious traffics than a single situation, which made the MTD defense results of each organization converge to the stable value quickly. Similarly, we judged the defense effect of an organization by the number of steps in each episode and the defender's benefits. We set up five situations to study the number of steps in each episode and the defender's benefits, settings are given in Settings #2.1-#2.5. The evaluation results are shown in Fig. 10. In addition, we set up eight situations to analyze the correlation between node quantity and defender benefits, settings are given in Settings #3.1-#3.8. The first four situations explored the impact of increasing node diversity within organizations while maintaining a constant number of organizations. The remaining situations

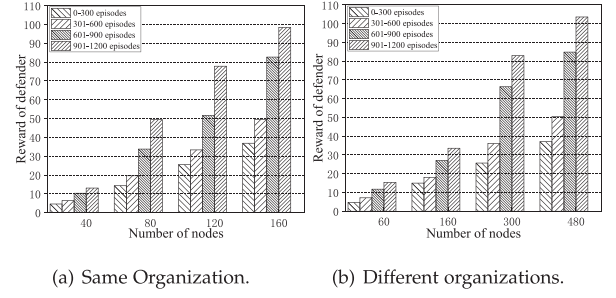


Fig. 11. Reward of defender with the different number of nodes.

considered the impact of increasing both the number of organizations and the number of nodes within the organization on defense effectiveness. The evaluation results are shown in Fig. 11.

Average value of interception rate for each node is denoted by α' , which is shown in (23).

$$\alpha' = \frac{1}{n * m} \sum_{i=1}^m \sum_{j=1}^m \alpha'_{ij} \quad 1 \leq i \leq m, 1 \leq j \leq n \quad (23)$$

Next, we evaluated changes of the expected value of α_{ij} (defined as α) and α' in blockchain node (in Section II-B). The relationship between α and α' reflected DDoS duration time to victims in organization. According to Section II-B, the sooner α was greater than α' ($\alpha \geq \alpha'$), the better of DDoS defense effect. See Settings #2.1, #2.3, and #2.5.

Finally, evaluations also considered the case of multiple organizations. We set MTD action of defenders as a defense round, the number of total saved nodes was the evaluation measurement. The evaluations set three groups for different m s (Settings #2.1, #2.3, and #2.5), see Fig. 12.

B. Experiment Results

Fig. 8 depicted that the increment of episodes had a negative relationship with learning steps, which implied that it was quicker for the defender to terminate DDoS attack at the end of DQN learning than at the beginning of DQN learning. The defenders could prevent DDoS attacks within 200 steps when an episode at 200. Efficiency was 96.22% higher than the situation when episode = 1. In addition, the number of steps and the attacker's profit were converged to 198 and -5 at 70 episodes, respectively.

In Fig. 9, the statistical data were the average value of the number of steps in the period of episodes. For example, the value in episode = 400 was the average number of steps from episode 200 to episode 400. Learning duration of bots=100 was less than 500 and 1000 number of bots when the episode was below 200. Except for the above situation, the learning duration was almost close in different scales of bots. In addition, the learning steps were stable at around 180 after 1000 episodes, which meant the defense model could converge to a high-efficiency state quickly.

Fig. 10 depicted that defenders received more rewards with increased collaboration among organizations. We evaluated the defense mechanism by setting all blockchain nodes under attack, using the number of execution rounds as the primary metric.

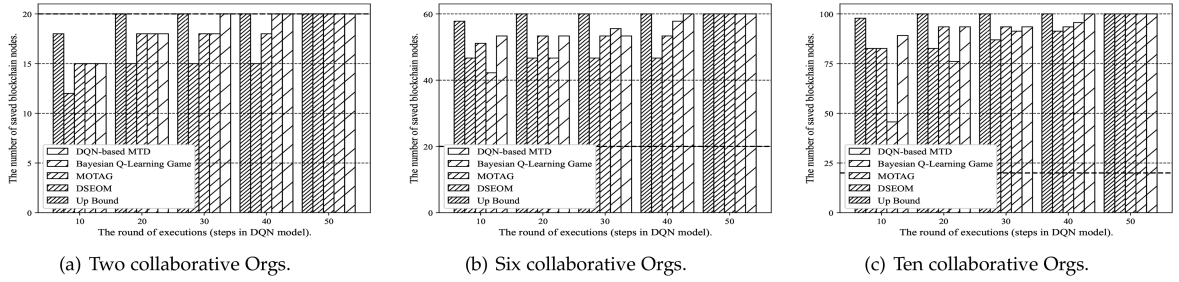


Fig. 12. The number of attacked nodes decreased as the number of defense MTD rounds increased. The total number of nodes was 20 in (a), 60 in (b), and 100 in (c).

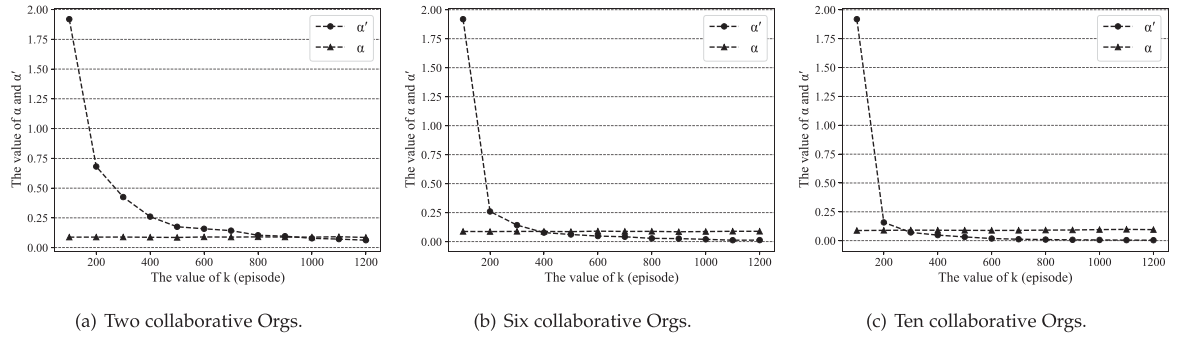


Fig. 13. Value of α and α' changed with episodes.

The results indicate that collaboration between multiple organizations improved defense efficiency. The defense effectiveness increased with the number of episodes. In addition, Fig. 11 depicted the defenders received more rewards with more nodes. We found that it was difficult to fully leverage the advantages of collaborative defense when the number of collaborative nodes in two groups was at a lower-level. However, as the number of nodes increased, there arose a problem of diminishing returns in benefits. Compared to the previous stage, the growth rate of benefits decreased from 240% to 40% and from 140% to 30%. Only an appropriate number of nodes could effectively exploit the defensive advantage. The evaluation results depicted that the increase in the number of nodes enhanced the defense reward; the defense effect also increased with increase of the number of organizations.

We extracted the accumulated data by ten rounds of defense MTD in Fig. 12. The data of Up Bound was the theoretical optimal approach for comparing other approaches. The results in Fig. 12 showed the proposed DQN-based could successfully filter all malicious traffics and the efficiency could be upon 94.95% of the up bound. In addition, considering the efficiency, the proposed approach saved all blockchain nodes faster than other approaches and the efficiency of the proposed approach performed well when a large scale of defenders were employed. The results demonstrated that the proposed approach could efficiently defend DDoS attacks in consortium blockchain. In the evaluation of α and α' , condition of α equal to α' was arrived about episode = 900, episode = 400 and episode = 300 in Fig. 13(a), (b), and (c) respectively. The three evaluation results intuitively reflected the increase of the number of organizations

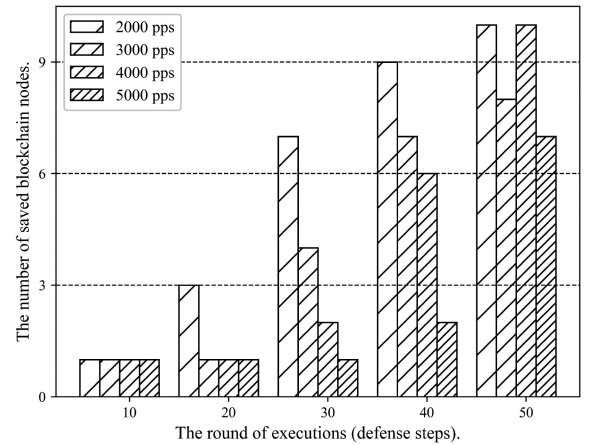


Fig. 14. The number of saved nodes in episode 1000.

can help victims terminate DDoS attacks in less duration time. Episodes in Fig. 13(c) was 66.67% fewer than that of Fig. 13(a).

Fig. 14 depicted that the number of safe blockchain nodes gradually increased with more defense rounds. In each round, defenders could quickly save all nodes at low attack intensities, but as intensity increased, mitigation speed slowed, though most nodes were saved within 50 rounds. Overall, under an attack intensity of 2000 pps, the mitigation efficiency was 42.9%, 50%, and 150% higher than under intensities of 3000 pps, 4000 pps, and 5000 pps, respectively. Fig. 15 demonstrated that attackers failed to obtain maximum benefits when the frequency of sharing malicious traffic was high, such as once every 500 ms, as this concurrently lowered training efficiency. The attack return also

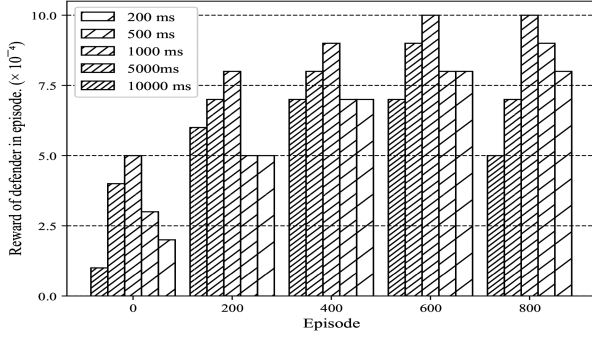


Fig. 15. Defender's reward in each episode with the different frequency of sharing DDoS messages (200 ms, 500 ms, 1000 ms, 5000 ms and 10000 ms).

decreased with sustained high sharing frequencies. Our experiments found that defenders achieved a maximum benefit when the malicious information-sharing frequency was approximately 1000 ms.

The results showed two advantages of our approach. One merit is that the efficiency of our DQN-based approach could be enhanced when the volume of the malicious traffics was stored. The other merit is that our approach could successfully filter the persistent malicious traffics by adopting our MTD method.

To measure the maximum rate at which consortium blockchain nodes process invalid transactions or empty blocks, we submitted a series of invalid transactions and generated empty blocks to the system, while monitoring and recording the speed at which each node processes these blocks. The experimental results showed that during our testing process, an average of 990 invalid transactions were intercepted per second, and the processing speed for empty block transactions was 17394 TPS.

In summary, in this work we assumed that rational attackers would adjust the optimal DDoS attack strategy to select suitable nodes and obtain the maximum profits in real-time when updating MTD strategies. A multi-stage game was formulated to abstract the attack-defend interactions, such that a Nash equilibrium would be reached. The experiment results had demonstrated that rational attackers would eventually terminate attacks when the attack profit became negative. The limitation of our method was that the accuracy of DDoS attack predictions still needed an improvement, which would be explored in our future work.

C. Security Analysis

In line with the threat model defined in Section II-A, we provided a brief security analysis in this section. The analysis would be given from the perspective of MTD that defended the proposed threats. There were four key MTD parameters involved in our experiment evaluations, which played vital roles in the defense.

On one hand, considering threats to executor nodes, we have proposed an MTD method supporting the moves of smart contract containers. The increment of the MTD rounds will decrease the total attack profit P_{total} . Results shown in Fig. 8 have depicted that the attack profit will be converged to a stable minimum value along with the growth of the training times, when the attack cost will be at a maximum point. In the following

Theorem V.1, we have demonstrated the effectiveness of our method to the threat.

Theorem V.1: Let the filtering capabilities of those nodes without deploying our defense method be α'_{ij} and those nodes using our defense method as α'_{ij} . There exist that the total filtering capability with adopting our defense method is greater than those of without adopting our defense method. The mathematical expression is given in (25).

$$\sum_{i=1}^m \sum_{j=1}^n \alpha'_{ij} > \sum_{i=1}^m \sum_{j=1}^n \alpha'_{ij} \quad 1 \leq i \leq m, 1 \leq j \leq n \quad (24)$$

Proof: Equation (25) is obtained from (4) showing the computation of the total filtering capability.

$$\sum_{i=1}^m \sum_{j=1}^n \alpha'_{ij} = \sum_{i=1}^m \sum_{j=1}^n \frac{a_{ij}^{pro}}{ka_{ij}^{cost}} \quad (25)$$

For an a_{ij}^{cost} , there is a value a_{ic}^{cost} , $c \in [1, n]$ making (26) existed.

$$\sum_{i=1}^m \frac{1}{ka_{ic}^{cost}} \sum_{j=1}^n a_{ij}^{pro} = \sum_{i=1}^m \sum_{j=1}^n \frac{a_{ij}^{pro}}{ka_{ij}^{cost}} \quad (26)$$

Deriving from (6), we can have following computation, shown in (27).

$$\begin{aligned} \sum_{i=1}^m \frac{1}{ka_{ic}^{cost}} V_i(s) &= \sum_{i=1}^m \frac{1}{ka_{ic}^{cost}} \left[\sum_{j=1}^n \alpha'_{ij} + \gamma V_i(s') \right] \\ &= \sum_{i=1}^m \frac{1}{ka_{ic}^{cost}} \sum_{j=1}^n \alpha'_{ij} + \sum_{i=1}^m \frac{1}{ka_{ic}^{cost}} \gamma V_i(s') \end{aligned} \quad (27)$$

Since $\sum_{i=1}^m \frac{1}{ka_{ic}^{cost}} \gamma V_i(s') > 0$, we can have a conclusion shown in (28) such that the Theorem V.1 has been proved.

$$\sum_{i=1}^m \sum_{j=1}^n \alpha'_{ij} > \sum_{i=1}^m \sum_{j=1}^n \alpha'_{ij} \quad 1 \leq i \leq m, 1 \leq j \leq n \quad (28)$$

□

On the other hand, our scheme moves logic nodes to reduce those threats to consensus nodes. Fig. 12 have depicted that our method could successfully decrease the number of the attacked nodes in a gradual manner. Our method allowed the consensus to remain functional even with a few malfunction nodes in the system. Fig. 12 exhibited that our method could guarantee a maximum amount of recovered nodes at each round of MTD moves; thus, consensus was primarily recovered. Theorem V.2 described the effectiveness of the defense method to the consensus threat.

Theorem V.2: Assume that the maximum amount of the malfunction nodes in a consensus is m' . The number of consensus nodes being attacked is v' . A certain time spot is denoted by t' . Thus, there exist a $v' < m'$ when $t > t'$.

Proof: Deriving from demonstrations in Section II-B, we can obtain (29).

$$a_{ij}^{pro} = a_{ij}^{be} - a_{ij}^{cost}$$

$$= (N_{ij}^{cont} + N_{ij}^{lon}) - (M_{ij}^{IP} + M_{ij}^{Port})$$

where, $1 \leq i \leq m, 1 \leq j \leq n$ (29)

According to demonstrations in Section IV, there exists a set $(N_{ij}^{cont}, N_{ij}^{lon}, M_{ij}^{IP}, M_{ij}^{Port})$, which makes $a_{ij}^{pro} < 0$, such that $v' \leftarrow (v' - 1)$. Thus, $\exists t'$ there exists $v' < m'$ when $t > t'$. Theorem V.2 has been proved. $\square$

VI. RELATED WORK

As a new research direction, defense of DDoS attack in blockchain has been explored in recent years. Perez et al. [35] leveraged Ethereum's metering mechanism and developed a genetic algorithm to generate low-throughput contracts that could gradually exhaust nodes' resources. Li et al. [36] proposed a DDoS method to Ethereum, called DETER, which used a low-cost DDoS rate to attack the txpool in Ethereum node and disrupted the normal processing of transactions. Compared with [36], the DETER attack mainly targeted the txpool of Ethereum node and also utilized the metering mechanism of Ethereum. On the basis of the huge cost of mining-based DDoS attacks [37], [38], [39], Mirkin et al. [40] proposed a scheme called BDoS, in which attackers are driven by external incentives to prevent the normal operation of the blockchain. With only small resources invested, BDoS can motivate other miners to stop mining.

As a severe threat using bots to send malicious traffics for blocking services or essential links, the internal clients were used for launching DDoS attacks and the identity authentication was bypassed in the consortium blockchain [10]. A group of DDoS defense methods has been explored by prior studies, such as MTD-based defense, filtering, SDN-based defense, collaboration-based methods, and reinforcement learning-based defense methods.

Even though DDoS threats to consortium blockchains are observable, constructing effective defense methods was still encountering challenges [26]. One of challenges was that nodes in consortium blockchain were internally public so that this setting provided attackers with attack chances by attacking at public nodes' addresses. In general, attack features are hardly to be detected when they derive from internal nodes [33].

The collaborative DDoS defense assembled resources from multiple defenders to detect/defend against DDoS attacks, in order to guarantee the system security in a large scale. Sivaramakrishnan et al. [20] proposed a method, called SENSS, to assist victims to request DDoS defense resources from ISPs. In another exploration, Daniel et al. [41] developed a distributed attack mitigation platform for mitigating amplification attacks and dropping additional malicious traffics. Similarly, a DDoS mitigation platform was proposed in [41] that exchanged DDoS rate information within internal nodes. The collaboration in [41] was designed among multiple defenders (e.g., Internet Exchange Points, IXP) rather than providing defense power for victims [20], which appeared a higher-level adaptability in multiple application scenarios in a large scale of the system. Rashidi et al. [17] proposed a NFV-based domain network, called CoFence, to construct a collaboration network for mitigating the large volume of DDoS attacks through resource sharing.

CoFence performed better in flexibility and hardware resource saving than [41] but brought an extra exchanging overhead. In our approach, the organizations were enabled to share malicious traffics with one another via implementing smart contracts in consortium blockchain, which meant that the historical attack records were stored in the distributed ledger, such that reliability could be ensured [17], [20], [41].

Reinforcement Learning (RL) was another option for facilitating DDoS attacks mitigation. Li et al. [42] proposed an RL-based DDoS mitigation in Internet of Vehicles (IoV), named FAST, that applied the Kalman filter and historical data analytic; however, this method was passive in analyzing and responding to the bots' actions. [43] has introduced RL-based MTD to proactively change IoV configuration for mitigating DDoS. Compared with [42], the dynamic defense environment proposed in [43] improved the DDoS defense efficiency. Rather than addressing efficiency, Zhou et al. [44], [45] proposed a shuffling method that used a multi-objective Markov decision process to model behaviors of attackers/defenders and balance resource costs and defense effects. Differing from prior work, our approach further synchronized the malicious traffics among multiple defenders to make RL implementations more efficient.

Furthermore, MTD is an alternative for mitigating DDoS attacks in consortium blockchain, due to its proactive defense mechanism and dynamic attacking strategy [44], [46]. In most cases, MTD relied on itself rather than using third-party tools. For instance, combining MTD with greedy shuffle algorithms had been examined in DDoS defense [33], [47], [48]. The mechanism implemented a greedy client-server nodes shuffle to continuously adjust connections of client-server nodes and distinguish malicious clients. Such a greedy-based shuffle algorithm had two drawbacks. On one hand, efficiency would be dramatically lowered down when a large number of nodes were involved, as operations were sequentially executed. On the other hand, this sort of methods had a lower performance in predictions due to lack of historical network states data.

SDN-based method also was explored for defending DDoS attacks, because of its flexibility and efficiency [22], [49]. One of features of SDN was decoupling network's data plane and control plane [50]. The work [22] designed a SDN-based defense method, called Bohatei, which combined SDN with NFV to lower down the defender's costs. Administrators could flexibly deploy a variety of different software on common servers, while NFV improved defenders' flexibility. Yang et al. [51] proposed a machine learning-based SDN solution to mitigating DDoS attacks. The method used *Support Vector Machine* (SVM) to detect malicious traffics, which represented a better robust than other DDoS detection methods, e.g., [22]. Unlike [51], Xu et al. [49] utilized the traffic monitoring function of SDN to detect DDoS attacks, which accurately found potential attackers as well as occupied servers. Another work [52] tried adopting a *Markov Decision Process* (MDP) in SDN-based solutions, by which the optimal traffic process strategies could be generated for preventing DDoS attacks.

In our approach, we assumed the attacker controls sufficient internal nodes to launch DDoS attacks, posing hazards equivalent to insider threats. Dissimilar to SDN-based DDoS defense methods, our approach lacks a centralized controller, instead

relying on adaptable defense actions at each organization. Our work explores a different dimension from others by emphasizing two key aspects. First, our approach enhances malicious traffic filtering efficiency through collaboration between defenders. This collaboration also improves the refinement performance of RL in detecting malicious traffic rates. Second, we introduce a moving defense strategy to adapt to diverse DDoS attacks from both external and internal bots, highlighting our approach's capability to defend against internal threats. Additionally, our approach leverages historical data by utilizing a DQN-based MTD method, which fully exploits records of prior DDoS attacks to enhance MTD performance.

VII. CONCLUSION

In this work, we had proposed a collaborative approach for mitigating DDoS attacks while emphasizing internal threats of consortium blockchain systems, through combining MTD with DQN. First of all, the collaborative working mode enabled multiple organizations to share data of malicious records within a trustworthy environment, thanks to the distributed ledger. The mode contributed to both training and efficiency of traffic filtering. Moreover, our approach improved the effectiveness of the MTD decision making due to the implementation of DQN. The decision was updated along with the state update of bots-defenders of each round. With the increasing records of the malicious traffics, we found that the efficiency of decision making could be enhanced. Our experiment evaluations had demonstrated that the proposed approach could successfully defend DDoS attacks by organizing multiple organizations in blockchain. The main finding was that the amount of participant organizations or episodes had a positive relationship with the defending performance.

In our future work, we will fully consider the impact of MTD's latency on DDoS defense and improve the performance of our defense approach while considering various constraints, such as timing limit. A scalable size of participant organizations will be explored in order to demonstrate the impact of a dynamic organization setting on defending varied types of DDoS attacks.

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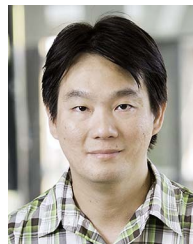
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